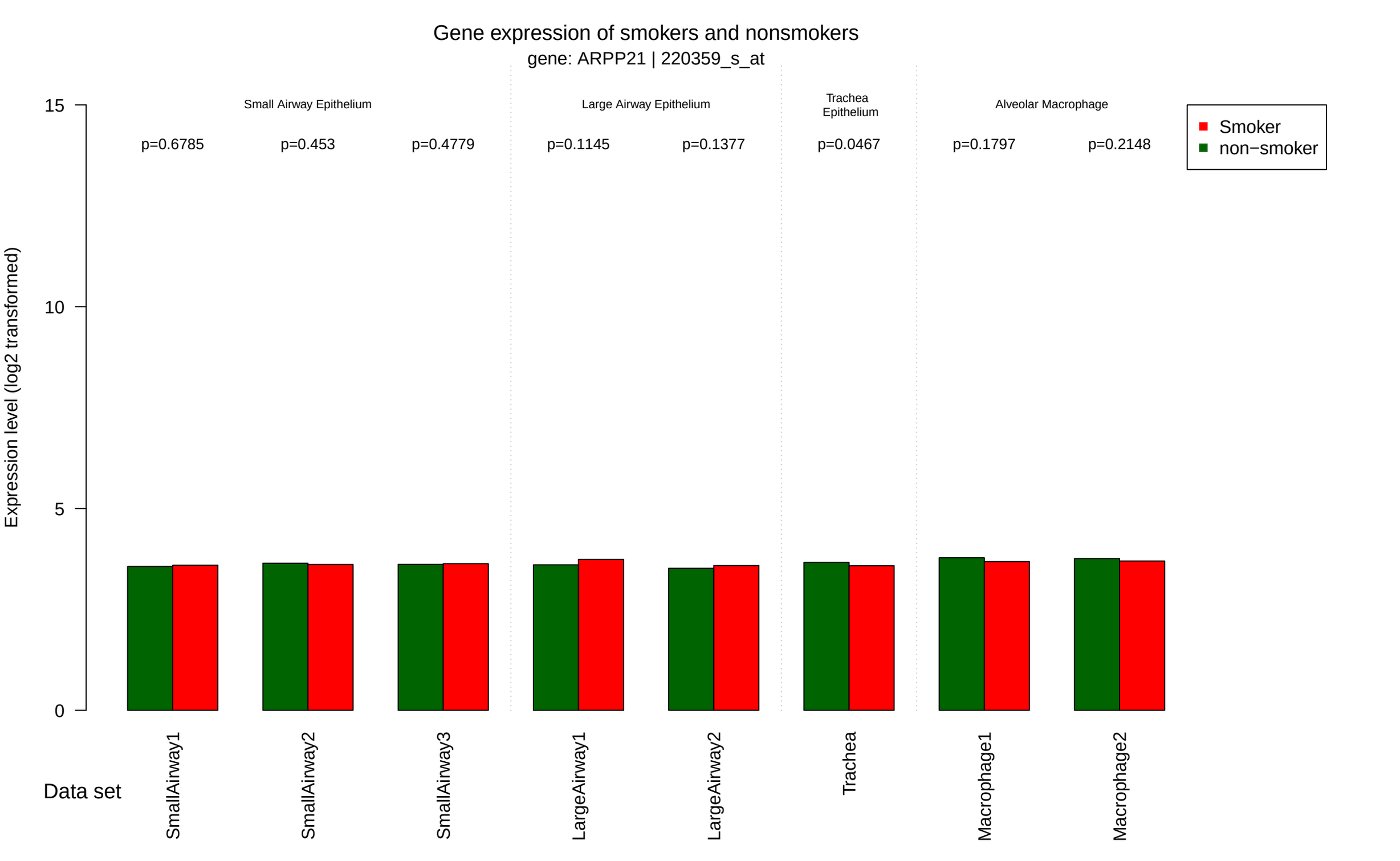
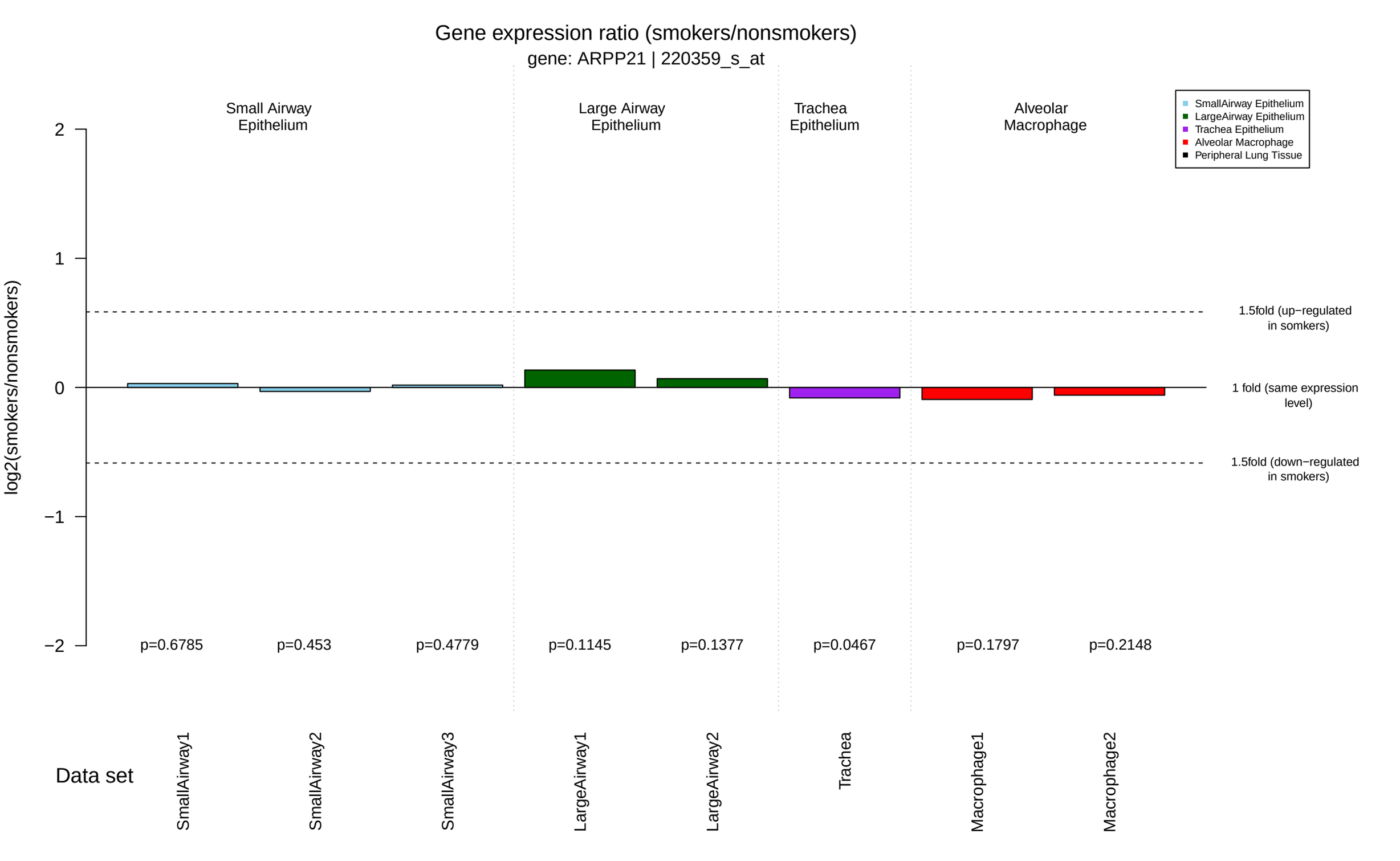
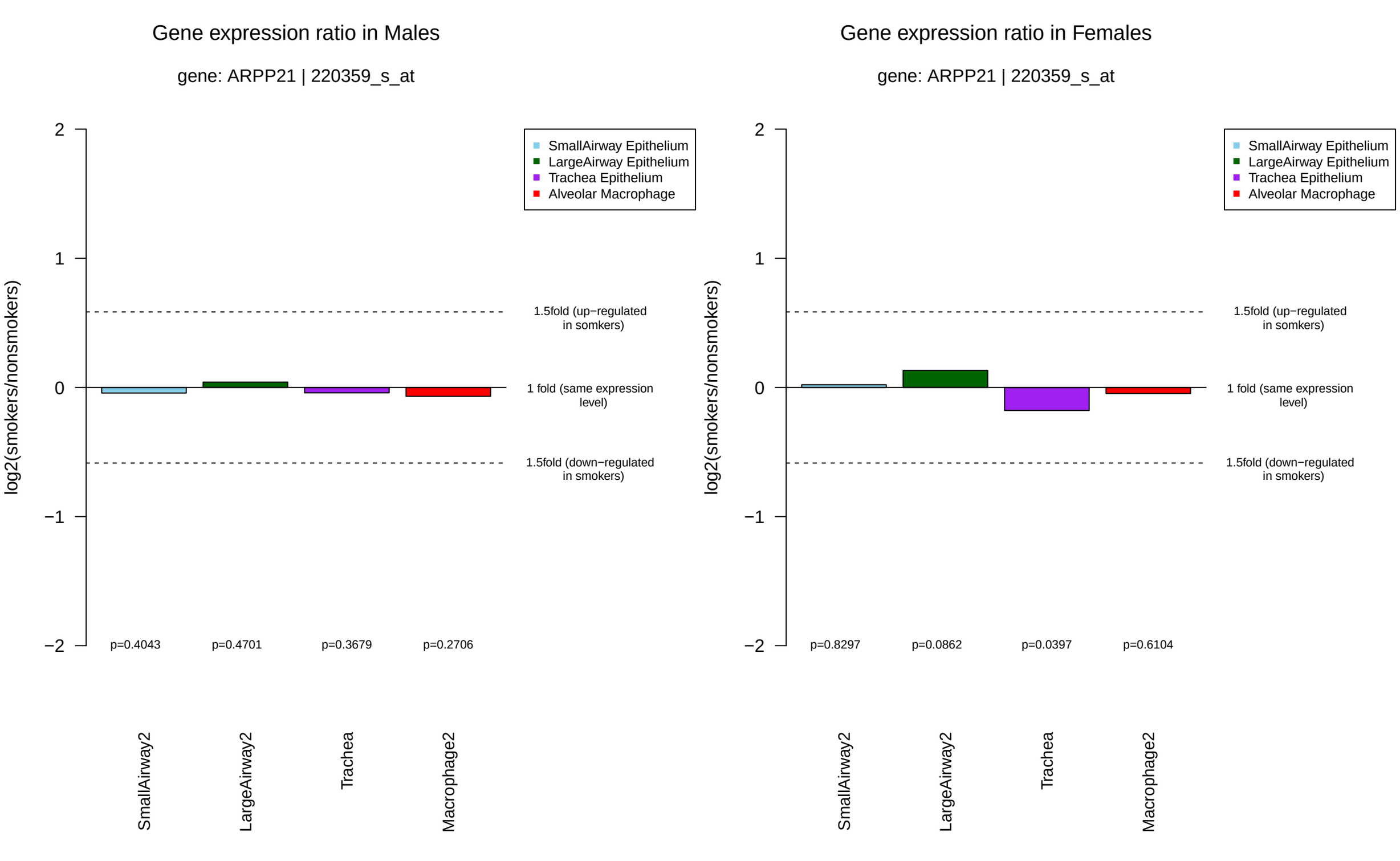
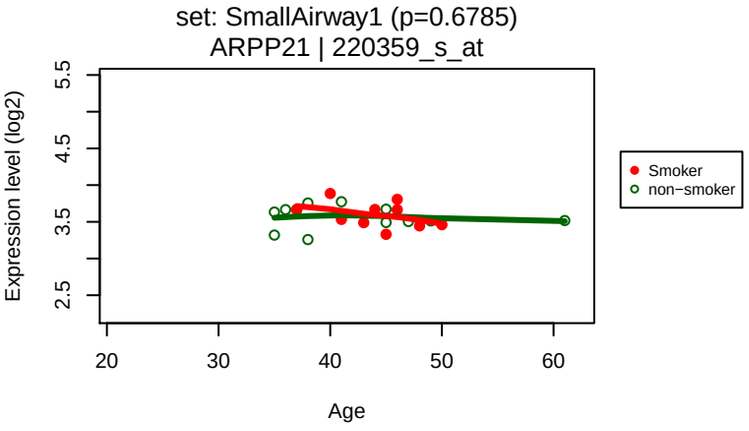
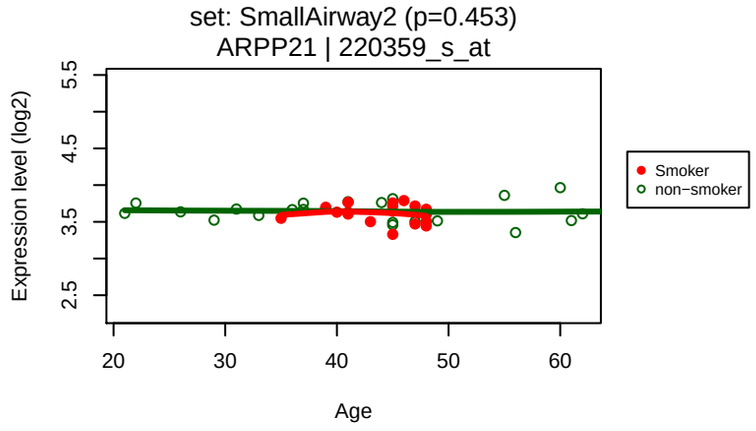
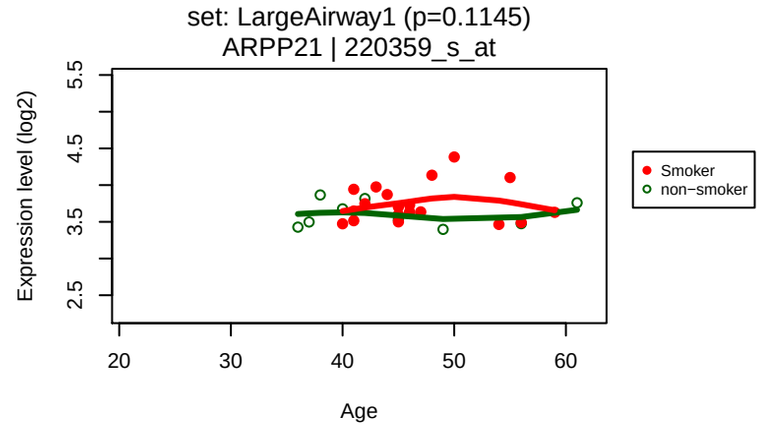

Gene expression of smokers and nonsmokers
gene: ARPP21 | 220359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6785
p=0.453
p=0.4779
p=0.1145
p=0.1377
p=0.0467
p=0.1797
p=0.2148
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

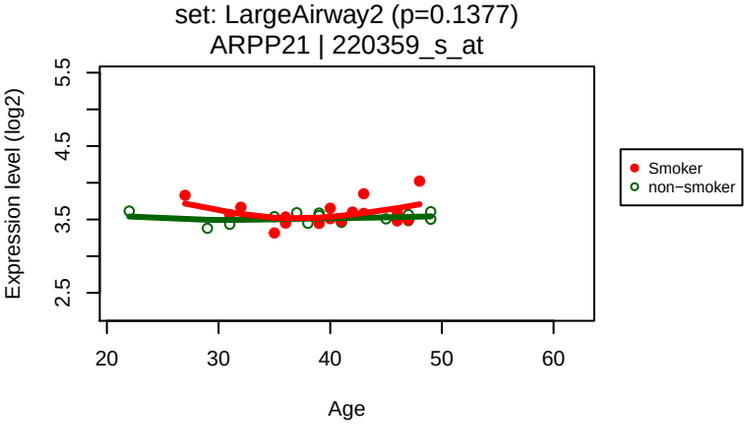

Gene expression ratio (smokers/nonsmokers)
gene: ARPP21 | 220359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6785
p=0.453
p=0.4779
p=0.1145
p=0.1377
p=0.0467
p=0.1797
p=0.2148
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

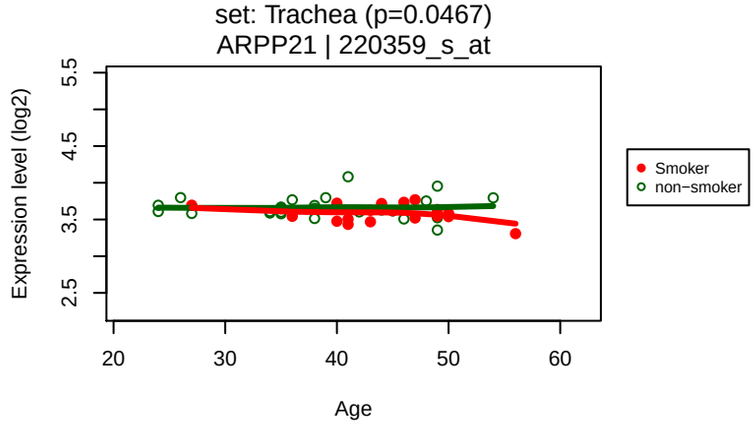

Gene expression ratio in Males
gene: ARPP21 | 220359_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4043
p=0.4701
p=0.3679
p=0.2706
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARPP21 | 220359_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8297
p=0.0862
p=0.0397
p=0.6104
SmallAirway2
LargeAirway2
Trachea
Macrophage2

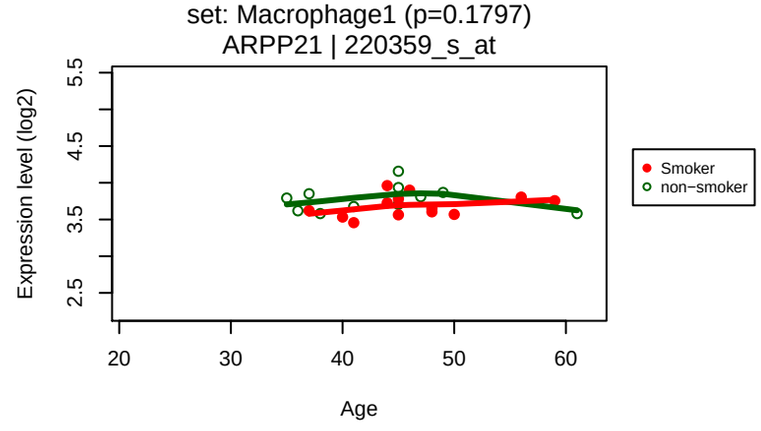

set: SmallAirway1 (p=0.6785)
ARPP21 | 220359_s_at
Expression level (log2)
Age
Smoker
non−smoker

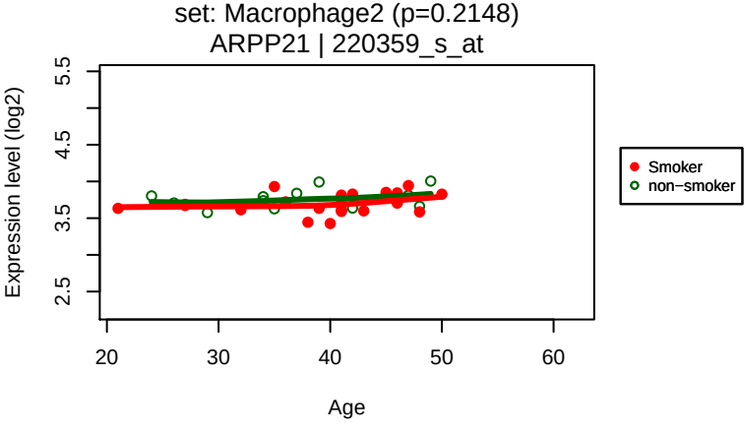

set: SmallAirway2 (p=0.453)
ARPP21 | 220359_s_at
Expression level (log2)
Age
Smoker
non−smoker

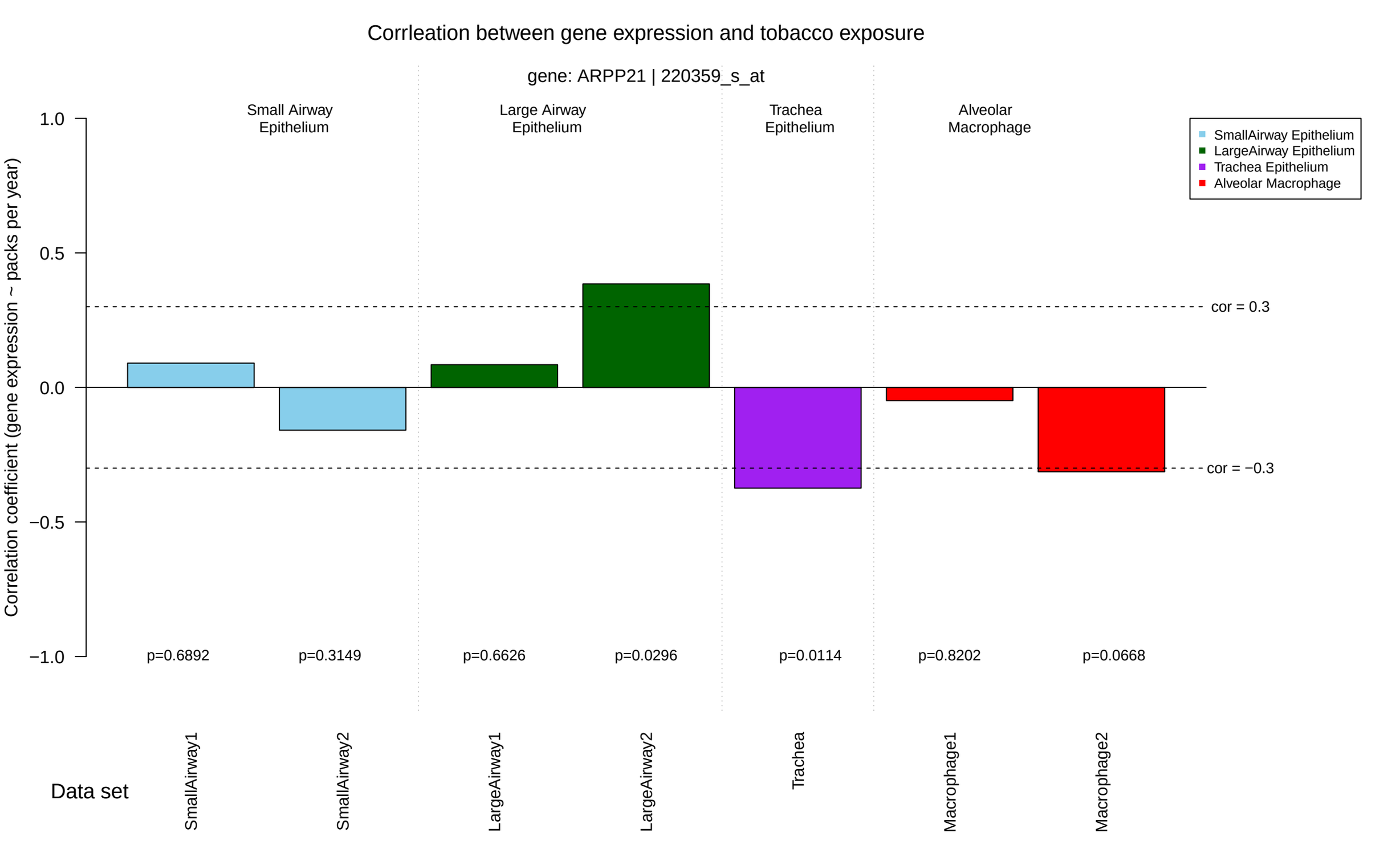

set: LargeAirway1 (p=0.1145)
ARPP21 | 220359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1377)
ARPP21 | 220359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0467)
ARPP21 | 220359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1797)
ARPP21 | 220359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2148)
ARPP21 | 220359_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARPP21 | 220359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6892
p=0.3149
p=0.6626
p=0.0296
p=0.0114
p=0.8202
p=0.0668
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2